Stochastic Processes 2021/22 Semester 1

Extension Questions 2

These questions are an extension to Tutorial 2, intended for personal revision only. Answers are available for all questions, but worked solutions are only available for some.

1. As in Tutorial 2 Question 1, let $\{S_n\}_{n\geq 0}$ be a simple random walk.

   (i) When $p = 2/3$ and $S_0 = 2$:
   (a) Calculate $\mathbf{P}(S_{10} \geq 0)$.
   (b) Find an approximation for $\mathbf{P}(S_{100} \geq 30)$.
   (c) Find $\mathbf{P}$(there exists $n > 0$ such that $S_n \leq -2$).

   Repeat (a)-(c) for the following cases:

   (ii) When $p = 3/4$ and $S_0 = 0$
   (iii) When $p = 1/3$ and $S_0 = -2$
   (iv) When $p = 1/2$ and $S_0 = -4$

2. A rival insurance company to the one in Tutorial 2 Question 2 has just started up with initial capital 2. Each year it receives premium income of 1 and pays claims. The total claim paid in each year i is Y_i where $\{Y_i\}_{i\geq 1}$ is a sequence of independent identically distributed random variables, each with the following distribution:

$$\mathbf{P}(Y_i = 0) = \frac{1}{4}; \quad \mathbf{P}(Y_i = 1) = \frac{3}{4} - p; \quad \mathbf{P}(Y_i = 2) = p.$$

   for some value of p. Given that the probability the company ever runs out of capital is 0.5625, find the value of p.

3. Fiasco the Cat is sitting on the fifth rung of an infinitely long ladder. Each minute he stays where he is with probability $\frac{3}{4}$, jumps up a rung with probability p, and jumps down a rung with probability q (such that $p + q = \frac{1}{4}$). Given that the probability that Fiasco eventually jumps off the bottom rung onto the floor is 0.03125, find p and q.

4. Another insurance company has initial capital 3. Each year it receives premium income of 1 and pays claims. The total claim paid in each year i is Y_i where $\{Y_i\}_{i\geq 1}$ is a sequence of independent identically distributed random variables, each with the following distribution:

$$\mathbf{P}(Y_i = 0) = \frac{7}{39}; \quad \mathbf{P}(Y_i = 1) = \frac{2}{3}; \quad \mathbf{P}(Y_i = 2) = \frac{1}{13}; \quad \mathbf{P}(Y_i = 3) = \frac{1}{13}$$

   (i) By finding an appropriate **lower bound**, show that the probability of this company running out of capital is more than the probability of the company in Tutorial 2 Question 3 (*Recall that the probability of the company in Tutorial 2 Question 3 ever running out of capital is* **at most** $(7/8)^5$.)

(ii) Let π_{20} denote the probability that the company runs out of capital within the next 20 years. In 200 simulations of this process, the number of times the company was ruined in 20 years was 87.

(a) Estimate π_{20}.

(b) Calculate an approximate standard deviation of your estimate.

(c) Estimate the number of simulations you would need to carry out for the standard deviation of your estimate to be less than 1% of the value of π_{20}

5. Alice invites Bob to play the following game:

- A marker is placed on a line of 10 boxes, labelled 1 to 10. The marker is placed in box 5 to start.
- Each turn Bob pays 10p to Alice and flips a fair coin.
- If the coin lands heads up the marker is moved up one space, if it lands tails up the marker is moved down one place.
- The games ends when the marker is placed in box 1 or box 10. If it ended in box 10, Alice pays Bob £6.

How much money will Bob expect to win or lose? (Hint: Condition on the next turn)

